



Dementia Prevalence Across Texas Counties

and the Association with Minerals Found in Groundwater

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The Growing Burden of Dementia

Alzheimer's Disease and Related Dementias (ADRD) - An Escalating Public Health Crisis



- **Projected Cases**
 - Over 7 million Americans with ADRD
 - ~13 million projected by 2050
- **Life Expectancy Post-Diagnosis**
 - ~5 years after diagnosis
- **Prevalence of Alzheimer's Disease**
 - Alzheimer's = 60-80% of dementia



Does Lithium Intake Reduce ADRD Risk?



Searching for Lithium's Minimum Effective Level



Data Sources

1

Texas Water Development Board (TWDB)

Measured mineral concentrations from wells across Texas (1919-2019), focusing on 13 minerals including lithium ($\mu\text{g/L}$, dissolved).

2

NORC Dementia DataHub

Medicare claims data for county-level dementia prevalence (under age 75) in 2020.



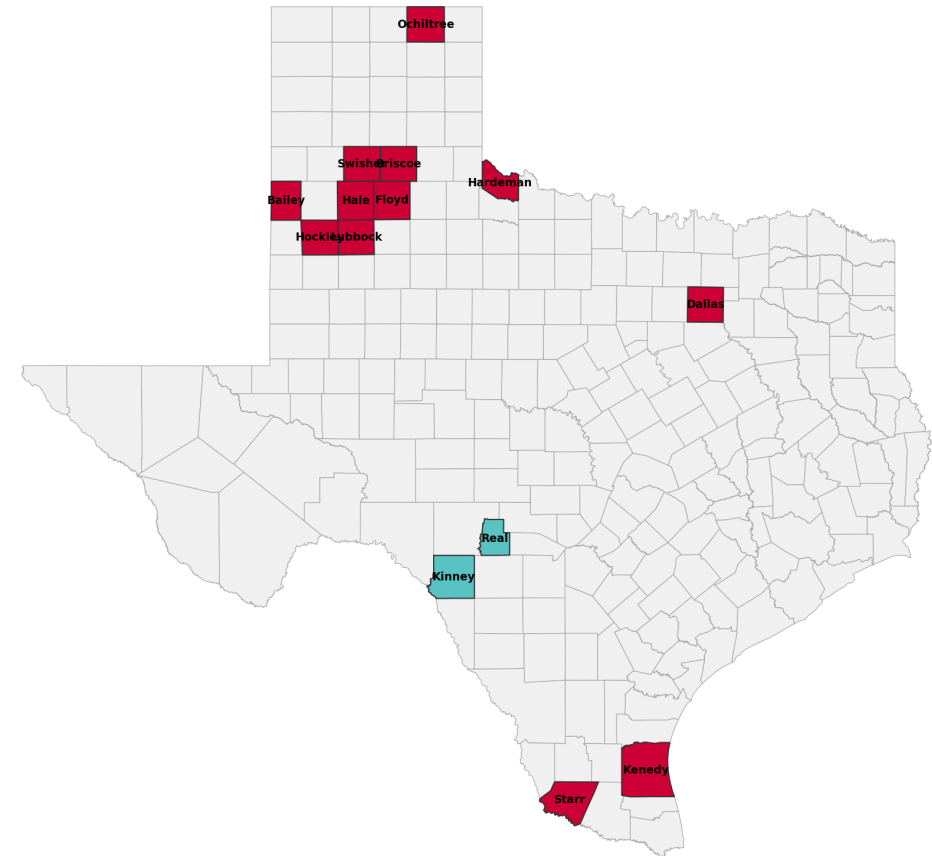
✱NORC | Dementia DataHub



Methods

- County-level ecological analysis
- **Key Challenge:** Exposure misclassification
 - Spatial averaging across diverse groundwater sources within counties.
- **Solution:** Exposure homogeneity classification
 - Identified counties with consistently high or low lithium levels in groundwater.
 - Resulted in a subsample of 14 homogenous counties.
- **Modeling Approach**
 - Ordinary Least Squares Regression (OLS) applied to:
 - The full Texas county sample.
 - The homogenous county subset.

Homogeneous Lithium Exposure Counties — Texas (Counties used in primary regression analysis)

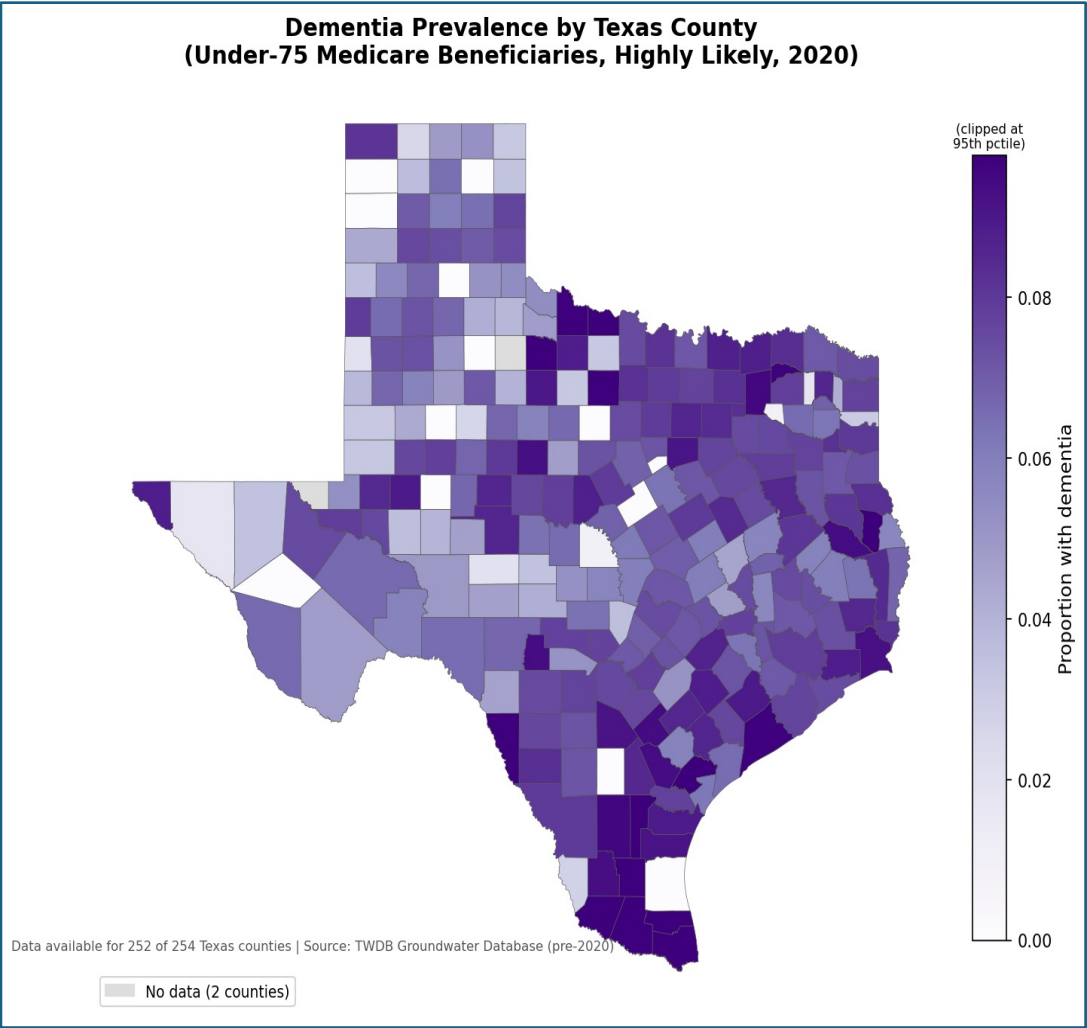
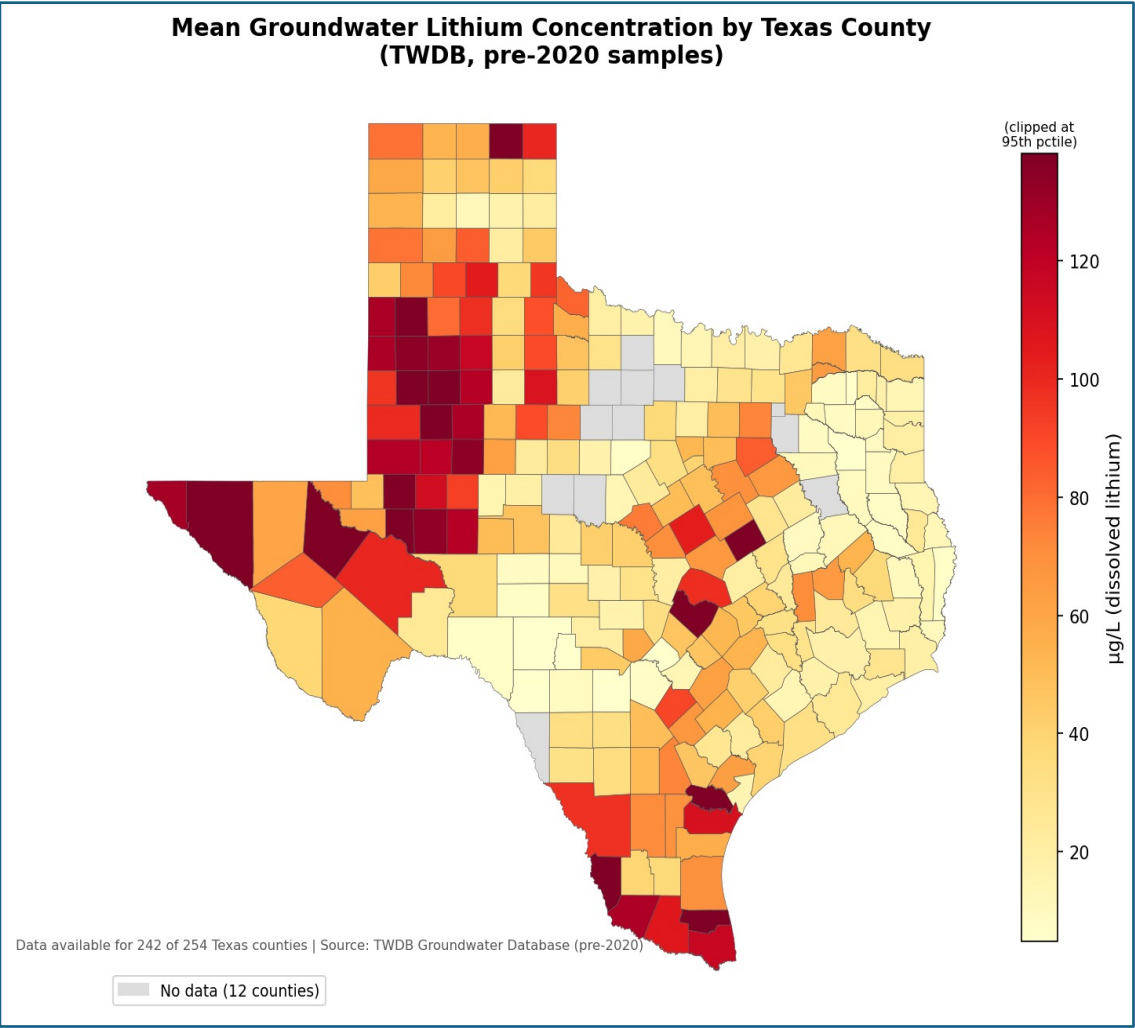


Key Innovation: Exposure Homogeneity Classification



Groundwater Lithium Concentration (µg/L)

County Dementia Prevalence (2020)



Geographic Patterns in Texas: Lithium and Dementia



Regression Table

Model	N	Li Coef	p-value	R ²	Interpretation
A — Full: dementia ~ Li	240	-0.000042	0.21	0.009	Null — expected under misclassification
B — Full: dementia ~ Li + Na	240	-0.000123	0.056	0.082	Near-significant inverse association
★ C — Homogeneous: dementia ~ Li + Na	14	-0.000112	0.73	0.226	PRIMARY RESULT — protective direction

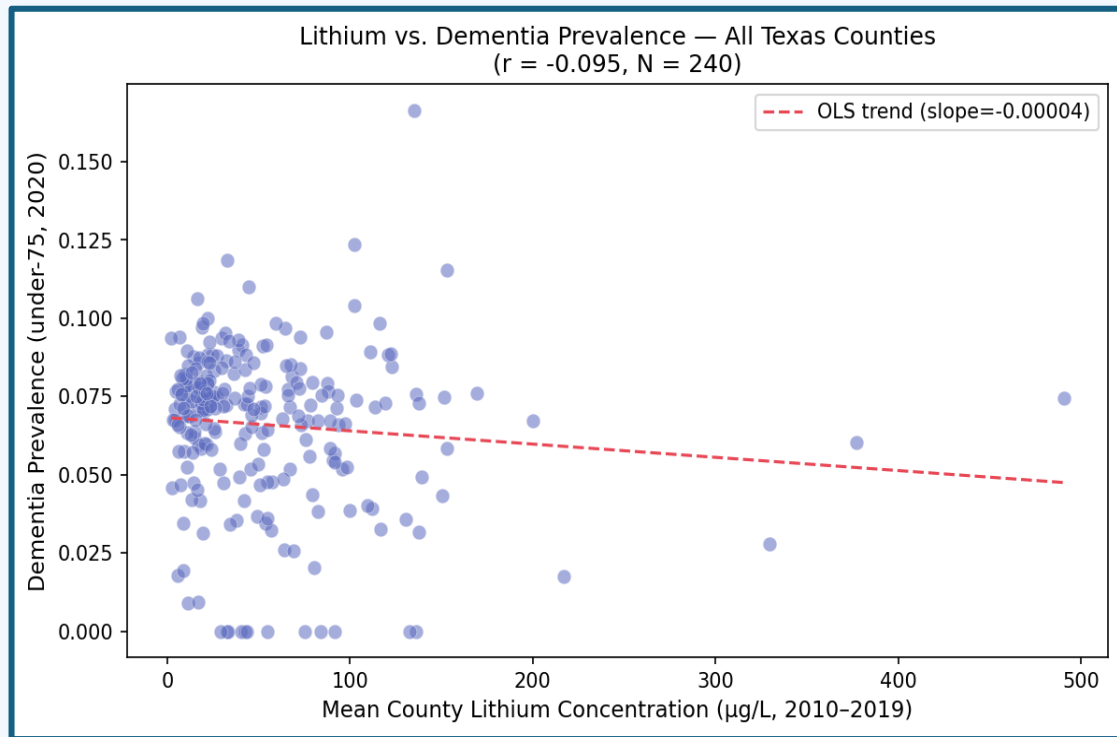
Blue shading = Full sample (N=240)

★ = Primary result

Coefficient direction remains protective

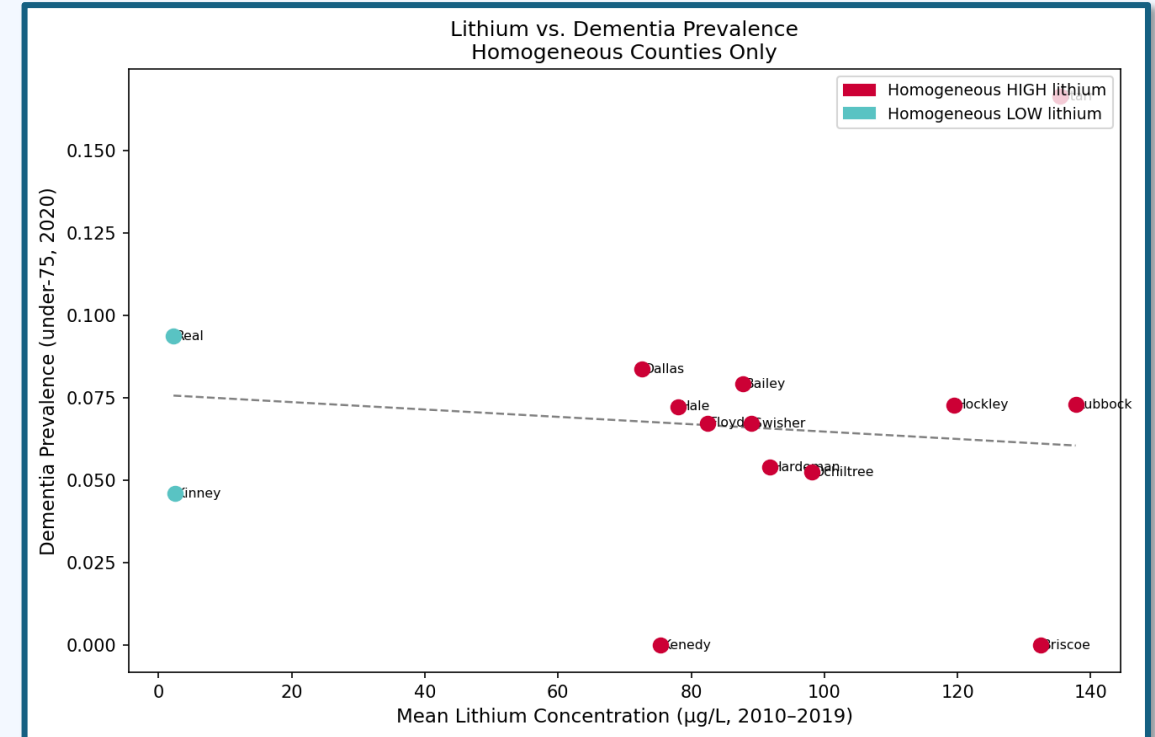


Results



All 240 Counties

- A weak, non-significant negative association was found between lithium and ADRD.
- This association became marginally significant when controlling for sodium ($p=0.056$).

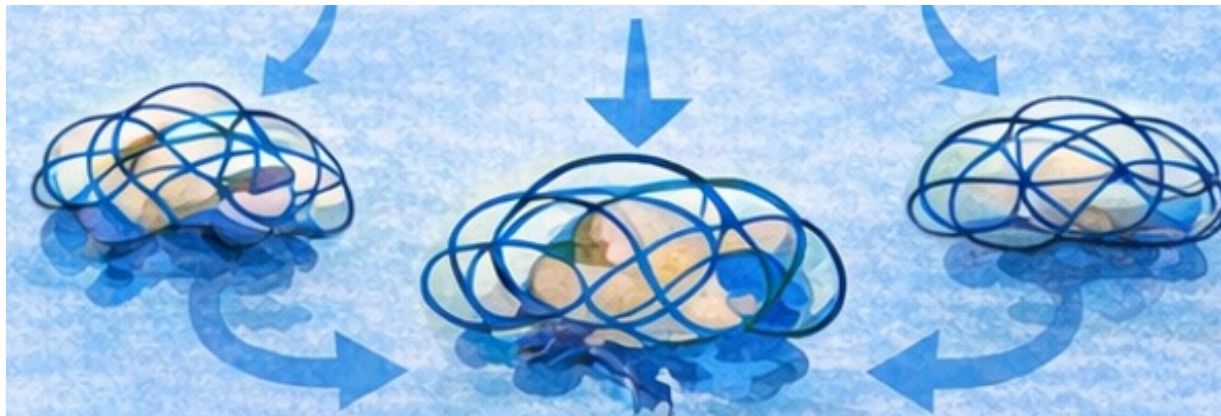
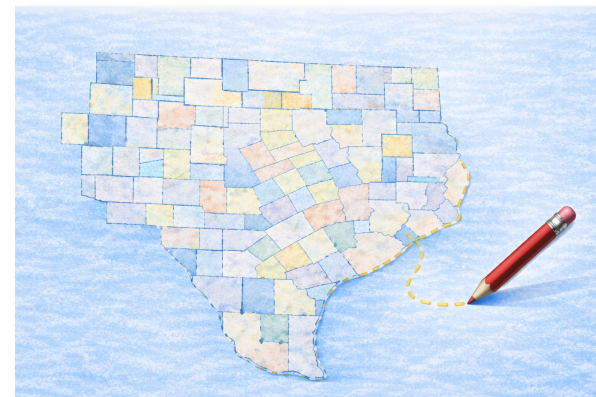


Only the 14 Counties with a Uniform Lithium Distribution

- As lithium increases, dementia decreases. Though it is a very weak association.
- The R^2 value increased significantly, indicating reduced exposure misclassification.



Limitations and Future Research



Questions & Feedback Welcome

We welcome any questions or feedback you may have.



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Thank You!

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